

Validating AI-generated classroom observations: Reliability, accuracy, and limits of LLM-based pedagogical judgment

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ARTICLE INFO

Keywords:

Classroom observation
Large language models
AI-Supported feedback
Reliability and validity

ABSTRACT

This study examines the reliability and accuracy of large language models (LLMs) for automated classroom observation using the World Bank's TEACH Primary framework. As education systems increasingly explore AI-based tools to scale teacher feedback and professional development, empirical validation of these systems is critical. Using a corpus of 12 primary classroom videos, we compared 8618 AI-generated evaluations from eight LLM endpoints against consensus-based ratings from certified TEACH experts. To account for model stochasticity, each model produced 10 independent evaluations per video–element pair. Reliability was assessed using variability and inter-rater consistency indicators, while accuracy was evaluated using error-based and concordance-based agreement measures. Results show substantial stochastic variability across repeated evaluations, with no model achieving uniformly high reliability across instructional elements. Agreement with expert ratings remained moderate at best. Importantly, reliability and accuracy did not co-vary systematically: models producing more stable scores did not necessarily align better with expert judgments, and models with stronger expert agreement often exhibited higher internal variability. In an exploratory analysis of model justifications, patterns suggest that LLMs tend to prioritize explicit verbal cues over contextual or implicit pedagogical evidence when generating high-inference judgments. These findings highlight structural limitations of current text-based AI observation pipelines and demonstrate that automated classroom observation cannot be treated as a uniform capability. The study provides empirical evidence to inform the design and validation of AI-assisted observation systems that integrate pedagogical expertise, measurement constraints, and complementary human judgment.

1. Introduction

Classroom observation has become a central mechanism for improving instructional quality and supporting teacher professional development. Teachers are consistently identified as the most influential in-school factor affecting student learning (Hanushek & Rivkin, 2010; Maulana et al., 2023), and interventions targeting instructional practices tend to yield more reliable effects than those focused solely on inputs such as infrastructure or formal qualifications (Snijlsteit et al., 2016). At the core of these interventions lie classroom interactions, which function as the primary conduit through which teaching translates into learning (Kirk et al., 2022; OECD, 2021).

High-quality classroom observation depends on validated rubrics, representative sampling of instruction, and rigorous observer training and calibration (Bell et al., 2019; Praetorius & Charalambous, 2018). A

wide range of observation tools have been developed, spanning low-inference instruments that focus on directly observable behaviors—such as Classroom Observations of Student–Teacher Interactions (COSTI) (Gunn et al., 2021), Five-In-20 Classroom Observation Tool (FCOT) (Buechlein et al., 2025), and the Flanders Interaction Analysis (FIA) (Amidon, 1966)—to high-inference frameworks that require professional judgment to integrate multiple sources of evidence. Among these, the World Bank's TEACH Primary framework has gained prominence due to its international validation (Molina et al., 2019) and demonstrated associations with student learning gains (Molina et al., 2020). However, despite these strengths, TEACH—like other high-quality observation systems—relies on trained human observers, making large-scale implementation costly and logistically complex.

Recent advances in artificial intelligence have intensified interest in Automated Classroom Observation (ACO) as a potential response to

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<https://doi.org/10.1016/j.caeai.2026.100612>

Received 29 January 2026; Received in revised form 21 April 2026; Accepted 13 May 2026

Available online 15 May 2026

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these scalability constraints. In particular, large language models (LLMs) offer the ability to process classroom transcripts rapidly and at scale, automating aspects of instructional coding that have traditionally been labor-intensive (Wang, Gong, et al., 2025). Research indicates that LLMs can classify classroom dialogue substantially faster than human coders (Long et al., 2024) and can approximate expert judgments across a range of instructional frameworks, including Mathematical Quality Instruction (MQI) and Classroom Assessment Scoring System (CLASS) (Demszky et al., 2021; Demszky & Hill, 2023; Alic et al., 2022). From a professional development perspective, such systems hold promise for delivering timely, formative feedback that supports teacher reflection and iterative improvement, and have been proposed as diagnostic tools that allow educators to interrogate specific dimensions of their practice (Celik et al., 2022; Dogan et al., 2025; van Camp et al., 2025; Wang, Gong, et al., 2025; Demszky et al., 2021).

Despite this promise, the use of LLMs for classroom observation raises fundamental concerns about validity and reliability. Critics argue that automated scoring systems may rely on superficial textual features rather than substantive pedagogical content (Bennett & Zhang, 2015, pp. 128–141). Text-based approaches inherently omit multimodal cues—such as intonation, facial expressions, and gestures—that are central to classroom interaction (Hou et al., 2024; Saini et al., 2025; Atal et al., 2023). Beyond these limitations, generative models introduce an additional challenge: stochasticity, that is, the tendency of these models to produce different outputs from the same input due to inherent randomness. Unlike human observers, whose reliability can be strengthened through calibration, LLMs may produce meaningfully different scores for the same lesson under identical conditions (Bauer et al., 2025; Seo et al., 2025). While a growing body of work has examined the accuracy of AI-generated scores relative to expert benchmarks, far less is known about their stability across repeated evaluations in educational contexts.

This study responds to this gap by systematically examining both the reliability and accuracy of LLM-based classroom observation using the TEACH Primary framework. Rather than treating automated observation as a monolithic capability, we analyze how model outputs vary across repeated identical evaluations and across instructional elements, and we identify conditions under which misalignment with expert judgment is most pronounced. By doing so, the study explores potential risks associated with deploying AI-assisted classroom observation systems, particularly in contexts where LLMs may be used directly by teachers or school leaders as evaluative tools without expert mediation. These risks extend beyond technical performance, highlighting the need to move toward approaches that incorporate expert knowledge modeling and careful evaluation pipeline design. The present study takes an initial step toward addressing these uncertainties through a structured empirical analysis.

The study makes three interrelated contributions to the literature on AI-assisted classroom observation. First, it provides empirical evidence on intra-model variability by systematically examining the stability of LLM-generated scores under repeated, identical conditions—an aspect of automated observation that has been largely overlooked in prior research. Second, it offers a comparative analysis of eight state-of-the-art LLM endpoints against consensus-based ratings from certified TEACH expert using a high-inference instructional framework, extending existing validation work beyond single-model or single-pass evaluations. Third, through element-level analyses, the study identifies specific pedagogical dimensions in which automated observation systematically underperforms, thereby clarifying the instructional and measurement conditions under which AI-generated feedback is more or less defensible. Together, these contributions advance a more nuanced understanding of how automated classroom observation can be responsibly integrated into teacher professional development systems.

2. Related work

2.1. Automated classroom observation

Classroom observation has long been recognized as a central mechanism for improving instructional quality and supporting teacher professional development. However, the cost, time, and training demands associated with human observation have limited the scalability of even the most rigorously validated frameworks. In response, researchers have increasingly explored Automated Classroom Observation (ACO) systems that leverage advances in natural language processing and machine learning to analyze classroom interactions.

Early ACO efforts focused on low-inference coding schemes that prioritize directly observable behaviors and interaction patterns. These include instruments such as FIA (Amidon & Hough, 1967), which categorizes teacher–student talk moves, as well as more recent tools such as COSTI (Gunn et al., 2021) and FCOT (Buechlein et al., 2025), which emphasize structured coding of instructional behaviors from transcripts or video. Because these instruments rely on clearly defined, surface-level indicators, they have proven more amenable to automation and have served as important testbeds for early AI-based observation systems.

More recent work has extended automated observation to higher-inference instructional frameworks that aim to capture the quality of teaching rather than its frequency or form. Studies applying machine learning and language models to frameworks such as MQI and CLASS suggest that AI-based systems can approximate expert ratings under certain conditions (Demszky et al., 2021; Demszky & Hill, 2023; Alic et al., 2022). These approaches move beyond surface coding to engage with constructs such as instructional rigor, responsiveness, and classroom discourse quality, substantially expanding the scope of what automated observation systems attempt to measure.

Despite these advances, applying ACO to comprehensive, high-inference frameworks remains an open challenge. Frameworks such as the World Bank's TEACH Primary require observers to integrate multiple sources of evidence across a lesson segment and to apply explicit scoring rules that distinguish between the presence and absence of pedagogically meaningful practices. TEACH has been validated across diverse national contexts and shown to correlate with student learning outcomes (Molina et al., 2019, 2020), but its reliance on trained professional judgment makes it particularly demanding for automation. To date, relatively little work has examined how automated systems perform across individual TEACH elements or how closely their scoring behavior aligns with the rubric logic used by certified observers.

2.2. LLMs as evaluators (LLM-as-a-judge)

The emergence of large language models has expanded the scope of automated evaluation beyond rule-based or feature-engineered approaches. LLMs have been proposed as general-purpose evaluators capable of scoring complex artifacts, including essays, dialogue, and instructional interactions, directly from natural language input. This paradigm, often referred to as *LLM-as-a-judge*, has received growing attention in educational assessment and learning analytics, particularly for tasks involving complex, open-ended artifacts (Li et al., 2025).

In educational contexts, LLMs have been used to score written responses, generate formative feedback, and evaluate instructional dialogue at scale (Long et al., 2024; Wang, Gong, et al., 2025). Comparative studies suggest that, under certain conditions, LLM-generated scores can align moderately with expert judgments and reproduce some patterns of human evaluation. For example, Whitehill and LoCasale-Crouch (2024) applied LLMs to estimate CLASS Instructional Support scores from classroom transcripts, reporting correlations approaching human inter-rater reliability while also noting sensitivity to task formulation and modeling choices. These findings have fueled interest in the potential of LLMs as scalable evaluative agents in teaching and learning environments.

At the same time, prior work has raised open questions about the use of LLMs as evaluators. Scholars have questioned whether generative models rely on surface-level textual cues, learned instructional scripts, or other proxies that may not fully capture the constructs targeted by established assessment frameworks (Bennett & Zhang, 2015, pp. 128–141). Others have highlighted challenges associated with applying automated evaluation to rubric-driven tasks that require disciplined adherence to explicit criteria rather than holistic impressions (Williamson et al., 2012). Such concerns may be especially consequential in observational settings, where valid scoring depends on consistent application of rubric logic rather than global impressions of instructional quality. However, empirical examinations of how these issues manifest in complex, pedagogically grounded classroom observation frameworks remain limited.

2.3. Reliability, validity, and measurement theory in AI scoring

Beyond questions of accuracy, the deployment of AI-based evaluators raises fundamental measurement concerns related to reliability and validity. In human observation systems, reliability is established through structured training, calibration, and ongoing monitoring of inter-rater agreement. By contrast, LLMs are inherently stochastic, meaning that repeated evaluations of the same input may yield different outputs even under identical conditions.

While several studies have examined the accuracy of LLM-generated scores relative to human benchmarks, far fewer have investigated their stability across repetitions or over time. Emerging evidence suggests that intra-model variability can be substantial, particularly for high-inference judgments (Bauer et al., 2025; Seo et al., 2025). This variability poses a serious challenge for the use of LLMs in professional or evaluative contexts, as inconsistent scoring undermines both fairness and interpretability.

Measurement theory further underscores the importance of distinguishing between reliability and validity. A system may produce highly consistent scores that are nonetheless misaligned with the intended construct, or conversely, generate accurate scores that lack sufficient stability for practical use. For classroom observation, where feedback is intended to guide reflective practice and professional growth, both properties are essential. Yet the interaction between stochasticity, construct definition, and element-level operationalization remains underexplored.

By situating LLM-based classroom observation at the intersection of these literatures, the present study addresses a critical gap. It moves beyond single-pass accuracy evaluations to examine how reliability and validity jointly shape the defensibility of automated observation across different instructional elements.

2.4. Datafication, teacher receptivity, and LLMs in professional development

Beyond these measurement considerations, the deployment of AI-based observation systems introduces broader educational and socio-technical implications. The use of such systems is part of a wider process of educational datafication, understood as the transformation of classroom interactions into quantified, machine-readable data. This reliance on data influences decision-making processes across all levels of educational systems, from individual classrooms to national policy (Jarke & Breiter, 2019). In turn, this raises questions about what is measured and why, and in the case of teaching, about which forms of teaching and learning become legible or remain invisible when mediated through automated systems. For example, prior work suggests that increasing datafication and automation steers teacher–student interactions toward an objectified, measurement-driven dynamic, potentially reshaping the dialogical and relational dimensions of teaching (Mertala, 2024). In the context of classroom observation, the processing of transcripts through scoring pipelines foregrounds explicit and

textually recoverable features, while backgrounding embodied, tacit, and interactional dimensions of instructional quality. Understanding these implications of datafication complements technical validation studies in this domain.

Beyond technical performance, the practical value of AI-generated feedback depends on how teachers engage with and interpret such outputs. Research on AI-assisted professional development suggests that LLM-based systems hold promise for delivering timely, formative feedback that support teacher reflection (Celik et al., 2022; Dogan et al., 2025). However, this potential is contingent on teacher receptivity, on how teachers trust, critically interpret, and actively integrate AI-generated evaluations into their professional reasoning. Prior work has shown that teachers may engage with AI feedback in ways that are uncritical, selective, or resistant, depending on their professional agency, familiarity with the tool, and degree of expert mediation available (Wang, Gong, et al., 2025). When feedback is inconsistent, biased towards explicit evidence, or misaligned with teachers' perceptions or their own professional knowledge, the risk of distrust or misinterpretation increases. These dynamics underscore that the adoption of AI-generated classroom feedback is not merely a technical issue, but a professional and relational one, shaped by the conditions under which teachers encounter, interpret, and act on technology-driven recommendations.

The use of LLMs as tools for teacher professional development has attracted growing interest, largely due to their capacity to provide scalable, on-demand instructional feedback. The immediacy of AI-generated analysis may enable teachers to engage in reflective practice grounded in concrete instructional evidence rather than abstract theory, allowing them to revisit specific moments of their practice (van Camp et al., 2025). Emerging work also suggests that LLM-supported interventions can positively influence specific instructional practices (Wang, Gong, et al., 2025; Demszky et al., 2023). More broadly, LLMs have been proposed as diagnostic partners that allow educators to interrogate specific dimensions of their practice (Wang, Gong, et al., 2025). However, the pedagogical value of these systems remains conditional on their reliability and alignment with established professional standards. If AI-generated feedback is unstable across repetitions or systemically misaligned with expert judgement, its integration into professional learning cycles may introduce noise rather than support reflective growth. Empirical validation of these systems, including their accuracy and stability, is therefore not a post-deployment concern but a prerequisite for their responsible integration into teacher development frameworks.

3. Methods

3.1. Research questions

RQ1: To what extent do large language models (LLMs) produce stable classroom observation scores under identical prompting conditions?

RQ2: How accurately do LLM-generated scores align with consensus-based ratings from certified TEACH experts across TEACH Primary elements?

RQ3: Does model reliability and accuracy vary systematically across different pedagogical dimensions?

3.2. Research design

This study employed a quantitative, within-model observational design with expert benchmarking to examine the consistency and accuracy of large language models (LLMs) when evaluating teaching practices using a standardized classroom observation framework. Model-generated evaluations were systematically compared against consensus-based ratings from certified TEACH experts, which served as the reference benchmark.

The evaluation rubric used was the TEACH Primary protocol, an open and internationally validated instrument developed by the World Bank to assess teaching quality. TEACH was selected due to its open-access nature, the availability of extensive training resources, its adaptability to different educational contexts, and its multilingual documentation, all of which facilitate replication and methodological transparency. In practice, TEACH is used in both formative and summative contexts, supporting teacher professional development as well as institutional evaluation processes.

The TEACH Primary framework organizes classroom observation into three domains—Classroom Culture, Instruction, and Socioemotional Skills—which are operationalized into nine elements and twenty-eight observable teacher behaviors. During a standard TEACH observation, these behaviors are coded as low, medium, or high based on structured evidence gathered during two 15-min classroom segments. These qualitative behavior-level ratings are subsequently transformed into a five-point quantitative scale, which aligns with the numerical scoring required for the present analyses.

In this study, TEACH is used as a structured and expert-anchored measurement instrument to provide a reference point for evaluating LLM-generated scores. Accordingly, the research design focuses on the technical properties of automated observation—specifically, the reliability and accuracy of model outputs—rather than on the evaluation of teachers or the generation of feedback for professional development. This framing allows the analysis to isolate measurement-related questions, while leaving the implications for formative or summative applications as dependent on the validity of the underlying scoring process.

In this design, eight LLM endpoints were tasked with evaluating each classroom session across the nine TEACH elements. The selection of endpoints was intended to capture variability across the current landscape of state-of-the-art models at the time of the study, including differences in model families, sizes, and optimization strategies. In addition, the study focused on models accessible through publicly available APIs, reflecting realistic usage conditions in which teachers and practitioners interact with LLMs via commercial platforms rather than self-hosted systems requiring specialized infrastructure. By incorporating multiple endpoints from major providers under these conditions, the design enables a comparative assessment that goes beyond single-model evaluation and supports a more robust characterization of how LLM-based classroom observation behaves in practice.

To ensure consistent input and isolate model-specific behavior, the models were provided with verbatim transcripts of each classroom video rather than the videos themselves. This choice standardized the informational content available to the models and avoided confounds related to vision capabilities or multimodal processing, enabling a controlled examination focused exclusively on text-based pedagogical judgment. Further details on the selected models and their configurations are provided in Section 3.4.

To capture the intrinsic stochasticity of generative language models, each LLM produced ten independent evaluations for every video–element combination under identical prompting conditions. This repeated-measures structure enables estimation of score distributions and intra-model variability, rather than relying on a single model response. Given the discrete five-point TEACH scale, ten repetitions are sufficient to observe whether model outputs remain stable or fluctuate across categories, while keeping the total number of API calls within a feasible range. The study is not intended to produce highly precise performance estimates for individual models, but rather to examine general patterns of consistency and agreement across LLMs in classroom observation tasks. From this perspective, the selected number of repetitions provides a stable basis for comparing model behavior and identifying systematic sources of variability. This design supports the analyses described in Section 3.6.

3.3. Video corpus and expert evaluation

The dataset comprises transcripts derived from 12 classroom video recordings, each approximately 15 min in duration and consistent with the structure prescribed by the TEACH Primary observation protocol. All materials were drawn from official World Bank TEACH training resources used to prepare and calibrate human coders.

The videos originate from two national contexts, with eight recorded in English and four in Spanish. This cross-contextual composition introduces natural variation in instructional practices, discourse patterns, and classroom norms, while remaining aligned with TEACH's intended cross-cultural applicability.

All videos were transcribed verbatim, and each transcript was preserved in its original language. This decision was made to maintain linguistic and semantic fidelity and to replicate a realistic practitioner workflow in which a teacher records a lesson, generates an automated transcript using AI tools, and submits it to an LLM for feedback. Preserving the original language avoids confounds associated with translation and enhances ecological validity.

A reference evaluation was available for each video, derived from consensus among three certified TEACH experts as part of the official training materials. These evaluations are developed with a high level of rigor and are used in TEACH certification processes, ensuring detailed and carefully validated scoring. The experts evaluated the complete video recordings in accordance with the TEACH Primary guidelines, coding all 28 observable behaviors and their aggregation into nine elements. These ratings were subsequently mapped to the standardized five-point TEACH scale. For simplicity, we refer to these consensus-based evaluations as expert ratings throughout the paper.

An important aspect of the evaluation design is that expert ratings were based on full video recordings, whereas LLMs operated on verbatim transcripts. This difference reflects the typical conditions under which TEACH evaluations are conducted and the way AI tools are currently used in practice. The expert scores therefore represent the standard application of the TEACH framework, while the transcript-based condition for LLMs isolates their ability to approximate expert judgment from textual evidence alone. As a result, observed differences between expert and model evaluations should be interpreted in light of this informational gap, which forms part of the empirical context of the study rather than a confound to be eliminated.

The expert ratings serve as the ground truth for this study, providing a validated benchmark against which LLM-generated scores are compared and enabling systematic examination of variability across instructional dimensions.

3.4. LLM models and prompting procedure

Eight model endpoints representing five state-of-the-art LLM families were evaluated. For three families, two versions were included to capture performance differences related to model scale or update cycles. Table 1 summarizes the evaluated endpoints, including model families and pricing per one million input and output tokens. Prices correspond

Table 1

Overview of evaluated large language models and endpoint configurations. Prices reflect provider rates as of December 2025.

model endpoint	LLM family	input 1M tokens	output 1M tokens
gpt-4o-mini	GPT	US\$0.15	US\$0.60
gpt-5-mini	GPT	US\$0.25	US\$2.00
gemini-2.5-flash	Gemini	US\$0.30	US\$2.50
gemini-2.5-flash-lite	Gemini	US\$0.10	US\$0.40
claude-haiku-4-5-20251001	Claude	US\$1.00	US\$5.00
claude-sonnet-4-20250514	Claude	US\$3.00	US\$15.00
deepseek-reasoner-v3.2	DeepSeek	US\$0.28	US\$0.42
grok-4-0709	Grok	US\$3.00	US\$15.00

to December 2025, when all API calls were executed, and provide context on the relative market positioning and expected performance tiers of the models.

All interactions with the models were conducted via their respective APIs, ensuring controlled runtime conditions and reproducible request execution. Although decoding parameters such as temperature or top-p sampling could be customized, we retained the default settings for all models to reflect typical user interactions through standard chat interfaces. This choice preserves ecological validity and maintains each model's natural stochastic behavior.

For each video–element combination, a separate API call was executed, meaning that the nine TEACH elements were evaluated independently for each video. All calls were performed without conversation history, ensuring self-contained prompts and preventing contextual leakage across elements, videos, or repetitions. This design isolates element-level judgments and avoids confounds related to memory or prior responses.

Each prompt followed a standardized, multi-part structure designed to replicate a realistic expert-evaluation scenario and to provide the model with all necessary definitions from the TEACH framework. The prompt included, in order.

1. Role specification: The model was instructed to act as an expert TEACH Primary evaluator.
2. Task description: A concise explanation of the TEACH methodology and its scoring structure.
3. General scoring instructions: Guidance on how to justify ratings and how to interpret the 1–5 scale.
4. General rules: Explicit instructions to base the evaluation solely on the transcript, avoiding assumptions or external knowledge beyond the textual evidence.
5. Component-level contextual description: A brief introduction to the TEACH area (component) to which the element belongs (Classroom Culture, Instruction, or Socioemotional Skills) including its conceptual purpose within the TEACH framework. This contextualization helps the model situate the element within its broader pedagogical domain, consistent with TEACH's hierarchical structure.
6. Element-specific description: A textual definition of the TEACH element being evaluated.
7. Behavior-level rubric: A complete description of each of the behaviors associated with the element; The expected indicators for low, medium, and high performance; Frequently asked questions (FAQs) included in the TEACH manual to clarify ambiguous cases.
8. Class transcript: The verbatim transcript of the 15-min lesson, in its original language.
9. Response instructions: The model was required to (a) assign a score for each behavior with justification, and (b) assign a score for the element with justification.

All descriptions of elements, behaviors, rubric criteria, and FAQs were extracted verbatim from the official TEACH Primary Observer Manual (Spanish version), ensuring traceability and avoiding interpretive drift in the adaptation of TEACH materials for model prompting.

Following this structure, base prompts (excluding transcripts) ranged from 1528 to 3261 words, while transcripts added 428 to 1744 words depending on the video, resulting in large total input sizes per model call. Providing full rubric definitions alongside complete class transcripts ensured standardized and context-rich evaluation conditions across all models.

3.5. Data collection

For each of the 12 classroom videos, ten independent evaluations were generated for every TEACH element and model endpoint, resulting in a total of 8640 API calls (12 videos $\times$ 9 elements $\times$ 10 repetitions $\times$ 8 endpoints). To enable automated processing and consistent extraction of

model outputs, all prompts required responses to be formatted in JSON, with explicit fields for behavior-level ratings, element scores, and corresponding justifications.

During data collection, 32 responses did not conform to the required format or were incomplete (e.g., malformed JSON or missing fields). These outputs were excluded, yielding a final dataset of 8618 valid evaluations. No additional cleaning was required.

API calls for each model endpoint were executed consecutively. Due to provider latency, the longest interval between calls for a given endpoint was approximately one day. All evaluations for each endpoint were nonetheless collected within a sufficiently short window to minimize the likelihood of backend updates influencing model behavior.

3.6. Data analysis

The analysis focused on two complementary dimensions of LLM performance in the TEACH evaluation task: variability and accuracy. Variability was defined as the extent to which a model produces stable scores across repeated evaluations of the same video and element, while accuracy was defined as the degree of agreement between LLM-generated scores and expert ratings.

All analyses were conducted at the model-element level unless otherwise specified. Code was implemented in R using the packages *dplyr*, *psych*, and *DescTools*. Metrics were computed on the 8618 valid evaluations described above.

3.6.1. Variability (intra-model consistency)

Intra-model variability captures the extent to which an LLM produces consistent scores when evaluated repeatedly under identical conditions. Because LLMs rely on stochastic sampling, outputs may vary even when prompts and inputs remain unchanged, making variability a central indicator of reliability.

Variability was assessed using three complementary metrics: standard deviation (SD), Shannon entropy, and the intraclass correlation coefficient (ICC). SD captures numerical dispersion across repeated evaluations, entropy characterizes the distributional spread of scores across the five-point scale, and ICC(1,1), computed using a one-way random-effects model, estimates the proportion of variance attributable to systematic differences between videos relative to within-model fluctuations.

(a) Standard Deviation (SD)

For each model and element, SD was computed across the ten repeated evaluations per video. Low SD indicates stable scoring across repetitions, whereas high SD reflects inconsistent outputs that may reduce evaluative reliability.

(b) Shannon Entropy (H)

To complement SD, Shannon entropy was computed to quantify the uncertainty of score distributions across repetitions (Shannon, 1948; Cover & Thomas, 2006):

$$H = - \sum_{i=1}^k p_i \log_2(p_i)$$

where p_i is the empirical probability of assigning score i , and k is the number of observed categories. For a five-point scale, entropy ranges from 0 (all repetitions yield the same score) to $\log_2(5) \approx 2.322$ (uniform distribution). Unlike SD, entropy captures how dispersed or multimodal the score distribution is, making it sensitive to irregular response patterns. Empirically derived interpretive ranges for entropy levels are reported in Table 2.

(c) Intraclass Correlation Coefficient (ICC)

Table 2

Interpretation of shannon entropy for a five-point rating scale.

Entropy level	Entropy range (bits)	Interpretation	Justification for threshold
Low variability	0.00 – 0.80	The model produces highly consistent scores, typically concentrated in a single category with only occasional deviations.	Entropy below 0.80 arises in distributions where ≥ 70 –90% of repetitions fall into one category (e.g., $p = (0.8, 0.2, 0, 0, 0)$, $H \approx 0.72$), reflecting minimal unpredictability.
Moderate variability	0.80 – 1.40	The model distributes its scores across two or three categories, indicating variability but not full unpredictability.	This range corresponds to distributions with meaningful spread across a small subset of categories (e.g., $p = (0.6, 0.3, 0.1, 0, 0)$, $H \approx 1.15$), indicating partial stability with noticeable fluctuations.
High variability	> 1.40	The model shows inconsistent scoring across repetitions, spreading evaluations across many categories.	Entropy above 1.40 occurs when four or more categories are used with non-negligible frequency (e.g., $p = (0.4, 0.3, 0.2, 0.1, 0)$, $H \approx 1.85$), approaching the uniform maximum of $H = 2.322$ and indicating high randomness.

To evaluate consistency from a psychometric perspective, we computed the intraclass correlation coefficient ICC(1,1) using a one-way random-effects model. ICC is widely used in observational and educational research to assess the reliability of repeated measurements and is particularly suitable for settings where multiple ratings are produced for the same target.

In this study, each LLM generates ten independent ratings for the same video–element combination. These repeated evaluations can be interpreted as multiple “virtual raters” sampled from the model's internal stochastic process. ICC(1,1) quantifies the proportion of total score variance that is attributable to systematic differences between classroom videos, relative to variance introduced by random fluctuations across repeated model outputs.

The ICC(1,1) coefficient is defined as:

$$ICC(1,1) = \frac{MS_R - MS_E}{MS_R + (k-1)MS_E}$$

where MS_R is the between-video mean square, MS_E is the error (within-video) mean square, and $k = 10$ is the number of repeated evaluations per video.

Higher ICC values indicate that score variability is primarily driven by genuine differences between classroom videos rather than by instability in the model's outputs. Conversely, low ICC values indicate that random within-model variation dominates, suggesting limited reliability for evaluative use. Following Koo and Li (2016), ICC values below 0.50 indicate poor reliability, values between 0.50 and 0.75 moderate reliability, values between 0.75 and 0.90 good reliability, and values above 0.90 excellent reliability.

Unlike SD and entropy, which describe dispersion within a single video, ICC explicitly evaluates whether a model can consistently differentiate between instructional quality across multiple videos, making it particularly relevant for classroom observation scenarios.

3.6.2. Accuracy (agreement with expert)

Accuracy evaluates the degree of alignment between LLM-generated scores and those expert ratings, whose ratings are treated as the reference standard. Four complementary indicators were used: exact agreement (EA), mean absolute error (MAE), root mean squared error (RMSE), and Lin's concordance correlation coefficient (CCC). Together, these metrics capture both categorical agreement and continuous disagreement patterns between model and expert ratings.

(a) Exact Agreement (EA)

Exact Agreement represents the proportion of cases in which the model-assigned score exactly matches the expert rating. Given the discrete 1-5 TEACH scale, EA provides an intuitive measure of strict agreement and reflects how often the model reproduces expert judgments without tolerance for deviation.

(b) Mean Absolute Error (MAE)

MAE measures the average absolute difference between model and expert ratings:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_{model,i} - y_{expert,i}|$$

Because TEACH ratings are expressed on a 5-point scale, MAE directly quantifies disagreement in rating points. Lower values indicate closer alignment, while higher values reflect systematic or frequent deviations. MAE treats all errors equally and is robust to outliers.

(c) Root Mean Squared Error (RMSE)

RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{model,i} - y_{expert,i})^2}$$

Compared to MAE, RMSE places greater weight on larger discrepancies, making it sensitive to occasional high-magnitude errors. This is particularly relevant for evaluative tasks where large scoring deviations may have disproportionate practical consequences.

(d) Concordance Correlation Coefficient (CCC)

To assess overall agreement while accounting for both precision and systematic bias, we computed Lin's Concordance Correlation Coefficient (Lin, 1989):

$$CCC = \frac{2r\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2}$$

where r is the Pearson correlation between model and expert scores, σ_x^2 and σ_y^2 are their variances, and μ_x and μ_y are their means. CCC ranges from -1 to 1 , with higher values indicating stronger concordance and negative values reflecting systematic disagreement. CCC was not computed for model–element combinations where expert ratings showed zero variance, as concordance is undefined under such conditions. In these cases, MAE and RMSE were still reported.

4. Results

4.1. Intra-model variability

Table 3 summarizes the variability of model evaluations across all videos and TEACH elements using the three metrics described previously. The results reveal substantial differences between models in the dispersion and consistency of their repeated evaluations.

For the standard deviation metric, the mean SD varies notably across models, ranging from 0.20 for *claude-haiku-4-5-20251001*, the lowest value observed, to 0.52 for *gemini-2.5-flash*, the highest. Across all models, the minimum SD is 0, indicating that for at least one video-

Table 3

Summary of intra-model variability across all TEACH elements. “SD (mean)” is the average standard deviation across all video–element combinations. “SD (range)” reports the minimum and maximum SD values observed for each model. Entropy (mean) reflects the average Shannon entropy, and “% Low Entropy” indicates the proportion of combinations classified as low variability. “ICC (median)” is the median intraclass correlation coefficient across elements, and “N Elements ≥ 0.75 ICC” indicates the number of TEACH elements for which the model reached ‘good’ or ‘excellent’ reliability (ICC ≥ 0.75).

model	SD (mean)	SD (range)	Entropy (mean)	% Low Entropy	ICC (median)	N Elements ≥ 0.75 ICC
gpt-4o-mini	0.51	0 - 1.41	0.85	35.2%	0.70	2/9
gpt-5-mini	0.38	0 - 0.88	0.69	57.4%	0.65	2/9
gemini-2.5-flash	0.52	0 - 1.32	0.85	44.4%	0.64	2/9
gemini-2.5-flash-lite	0.42	0 - 1.16	0.74	52.8%	0.69	3/9
claude-haiku-4-5-20251001	0.20	0 - 0.74	0.36	76.9%	0.74	4/9
claude-sonnet-4-20250514	0.28	0 - 0.99	0.49	67.6%	0.69	3/9
deepseek-reasoner-v3.2	0.39	0 - 0.95	0.70	51.9%	0.53	1/9
grok-4-0709	0.36	0 - 1.05	0.63	53.7%	0.78	5/9

element combination the model produced identical scores across all ten repetitions. Maximum SD values differ substantially, reaching up to 1.41 for *gpt-4o-mini*, which reflects the widest dispersion recorded among the evaluated systems.

Entropy values show a similar pattern of variation. Mean entropy ranges from 0.36 for *claude-haiku-4-5-20251001* to 0.85 for *gpt-4o-mini* and *gemini-2.5-flash*. The proportion of model-video combinations categorized as “low entropy” also varies significantly, with *claude-haiku-4-5-20251001* exhibiting the highest proportion (76.9%) and *gpt-4o-mini* the lowest (35.2%).

The ICC values, summarized as medians across TEACH elements, show comparatively smaller variation. Median ICC values range from 0.53 for *deepseek-reasoner-v3.2* to 0.78 for *grok-4-0709*. When considering the number of TEACH elements for which models achieve “good” or “excellent” reliability (ICC ≥ 0.75), *grok-4-0709* reaches this threshold in 5 out of 9 elements, whereas *deepseek-reasoner-v3.2* reaches it for only 1 element.

Fig. 1 provides a visual summary of entropy values across models and TEACH elements. Consistent with the results reported in Table 2, *claude-haiku-4-5-20251001* exhibits generally lower entropy across elements, appearing lighter in most cells of the heatmap. A particularly notable case is *claude-sonnet-4-20250514* in the element *Perseverance*, which shows the lowest entropy among all model-element combinations. When examining variation across elements, the heatmap reveals substantial differences between models, with *Checks for Understanding* and *Positive*

Behavior Expectations displaying the largest concentration of high-entropy values (darker cells), indicating greater dispersion in model responses for these elements.

Table 4 summarizes variability metrics aggregated at the level of TEACH elements. The mean standard deviation is relatively similar across elements, ranging from 0.27 for *Social & Collaborative Skills* to 0.48 for *Checks for Understanding*, indicating modest differences in the overall dispersion of model evaluations. Entropy values show a comparable pattern, with the lowest mean entropy observed for *Social & Collaborative Skills* (0.47) and the highest for *Checks for Understanding* (0.85). The remaining seven elements fall within a narrower range between 0.54 and 0.77.

The proportion of model-video combinations categorized as “low entropy” also varies across elements. Perseverance shows the highest proportion at 65.6%, whereas Checks for Understanding shows the lowest at 36.5%, reflecting greater dispersion for this element. Median ICC values span a wider range, from 0.45 for Perseverance to 0.87 for Social & Collaborative Skills. Notably, Positive Behavior Expectations, Lesson Facilitation, and Perseverance have no models reaching the threshold for “good” reliability (ICC ≥ 0.75), whereas Social & Collaborative Skills is the only element for which all models achieve “good” or “excellent” ICC.

In summary, the variability metrics reveal clear differences across models and TEACH elements. Among the models, *claude-haiku-4-5-20251001* consistently shows the lowest dispersion, with the smallest

**Fig. 1.** Heatmap of mean Shannon entropy by model and TEACH element.

Table 4

Variability summary by TEACH element. “SD (mean)” corresponds to the average standard deviation across all model-video combinations for each element. “Entropy (mean)” reflects the average Shannon entropy for the same combinations, while “% Low Entropy” indicates the proportion of model-video pairs classified as low variability. “ICC (median)” represents the median reliability across models for each TEACH element, and “N Models ≥ 0.75 ICC” indicates the number of models that reached ‘good’ or ‘excellent’ reliability (ICC ≥ 0.75) for that TEACH element.

element	SD (mean)	SD (range)	Entropy (mean)	% Low Entropy	ICC (median)	N Models ≥ 0.75 ICC
Supportive Learning Environment	0.38	0.25 - 0.61	0.67	59.4%	0.76	4/8
Positive Behavior Expectations	0.44	0.18 - 0.72	0.77	46.9%	0.64	0
Lesson Facilitation	0.36	0.21 - 0.52	0.65	56.3%	0.52	0
Checks for Understanding	0.48	0.21 - 0.71	0.85	36.5%	0.71	2/8
Feedback	0.41	0.17 - 0.61	0.68	51.0%	0.59	1/8
Critical Thinking	0.39	0.15 - 0.53	0.56	59.4%	0.76	4/8
Autonomy	0.39	0.23 - 0.52	0.69	55.2%	0.72	3/8
Perseverance	0.31	0.05 - 0.44	0.54	65.6%	0.45	0
Social & Collaborative Skills	0.27	0.11 - 0.44	0.47	64.6%	0.87	8/8

mean SD, the lowest mean entropy, and the highest proportion of low-entropy combinations. At the other end of the spectrum, *gpt-4o-mini* and *gemini-2.5-flash* present the highest levels of dispersion across these metrics. When variability is examined from the perspective of TEACH elements, *Social & Collaborative Skills* presents the lowest variability, while *Checks for Understanding* displays the highest dispersion. The ICC values provide a complementary perspective: although *Perseverance* shows low SD and low entropy, its median ICC is the lowest among elements, indicating limited consistency in how models differentiate between videos for this element. Conversely, *Checks for Understanding*, despite showing higher within-video variability, achieves a higher median ICC, reflecting more stable relative scoring across videos. This contrast highlights that SD and entropy capture within-video stability, whereas ICC reflects the consistency of scoring patterns across videos for a given element.

4.2. Accuracy vs. expert ratings

Table 5 summarizes model accuracy relative to expert ratings using exact agreement (EA), MAE, RMSE, and the concordance correlation coefficient (CCC). Mean EA values range from 0.31 for *claude-haiku-4-5-20251001* to 0.55 for *grok-4-0709*, indicating substantial differences in the frequency of exact score matches across models. A similar pattern is observed for MAE, which ranges from 0.55 (*grok-4-0709*) to 0.95 (*claude-haiku-4-5-20251001*), with *gpt-4o-mini* showing a comparable value (0.94). RMSE values follow the same trend, spanning from 0.84 (*grok-4-0709*) to 1.31 (*gpt-4o-mini*), while *claude-haiku-4-5-20251001*

Table 5

EA (mean) indicates the proportion of exact matches with expert ratings. MAE (mean) and RMSE (mean) report average absolute and squared errors across videos and TEACH elements. CCC (median) summarizes typical model–expert concordance, while “N Elements with CCC < 0 ” indicates the number of elements with negative concordance. *Perseverance* is excluded from CCC due to zero variance in expert ratings.

model	EA (mean)	MAE (mean)	RMSE (mean)	CCC (median)	N Elements with CCC < 0
gpt-4o-mini	0.4	0.94	1.31	0.21	1/8
gpt-5-mini	0.53	0.62	0.94	0.23	3/8
gemini-2.5-flash	0.37	0.81	1.08	0.3	1/8
gemini-2.5-flash-lite	0.4	0.8	1.09	0.21	3/8
claude-haiku-4-5-20251001	0.31	0.95	1.18	0.16	1/8
claude-sonnet-4-20250514	0.47	0.7	0.98	0.28	2/8
deepseek-reasoner-v3.2	0.38	0.81	1.07	0.16	2/8
grok-4-0709	0.55	0.55	0.84	0.45	1/8

reaches 1.18.

The median CCC values also show clear differences across models. The lowest medians are observed for *claude-haiku-4-5-20251001* and *deepseek-reasoner-v3.2*, both at 0.16, while the highest corresponds to *grok-4-0709* with a median of 0.45. The second-highest value is obtained by *claude-sonnet-4-20250514* at 0.28, marking a substantial gap between the top-performing model and the rest. Negative CCC values are present in several cases: *gpt-5-mini* and *gemini-2.5-flash-lite* each show three elements with negative CCC, whereas four other models exhibit negative CCC for one element.

Fig. 2 provides a visual summary of CCC values across models and TEACH elements. As shown in the heatmap, *Critical Thinking* consistently exhibits higher CCC values across models, appearing lighter in most cells. In contrast, *Checks for Understanding* presents several darker cells, indicating lower agreement with the expert for multiple models. When examining performance by model, *grok-4-0709* displays relatively high CCC values across several elements, with the combination *Critical Thinking–grok-4-0709* reaching the highest CCC observed in the dataset (0.70). At the lower end of the scale, the combination *Checks for Understanding–gpt-4o-mini* shows the lowest CCC value, at -0.49 .

Table 6 presents accuracy metrics aggregated by TEACH element. Exact agreement (EA) varies substantially across instructional dimensions, ranging from 0.24 for Positive Behavior Expectations to 0.69 for *Perseverance*, indicating large differences in how frequently models reproduce expert ratings exactly. MAE values show a comparable pattern, spanning from 0.33 for *Perseverance* to 1.14 for *Positive Behavior Expectations*. Notably, the second-lowest MAE corresponds to *Lesson Facilitation* (0.64), highlighting a marked gap between the element with the smallest error and the remaining dimensions. RMSE values follow a similar trend, extending from 0.55 for *Perseverance* to 1.46 for *Checks for Understanding*.

Median CCC values also vary considerably. Two elements, *Positive Behavior Expectations* and *Checks for Understanding*, show negative median CCC values, indicating that model scores tend to diverge systematically from expert ratings for these dimensions. The highest median CCC is observed for *Critical Thinking* at 0.55. Regarding the proportion of negative CCC values, five elements exhibit 0 negative CCC across models, whereas both *Positive Behavior Expectations* and *Checks for Understanding* show negative CCC for six out of eight models, marking them as the elements with the largest concentration of disagreements with the expert ratings.

In summary, the accuracy metrics reveal substantial variation across both models and TEACH elements. Among the evaluated models, *grok-4-0709* consistently achieves the highest exact agreement and the lowest error values, together with the highest median CCC, whereas *claude-haiku-4-5-20251001* and *gpt-4o-mini* show lower EA and larger discrepancies relative to the expert ratings. From the element perspective, *Perseverance* exhibits the highest EA and the lowest error values, while *Positive Behavior Expectations* and *Checks for Understanding* display the

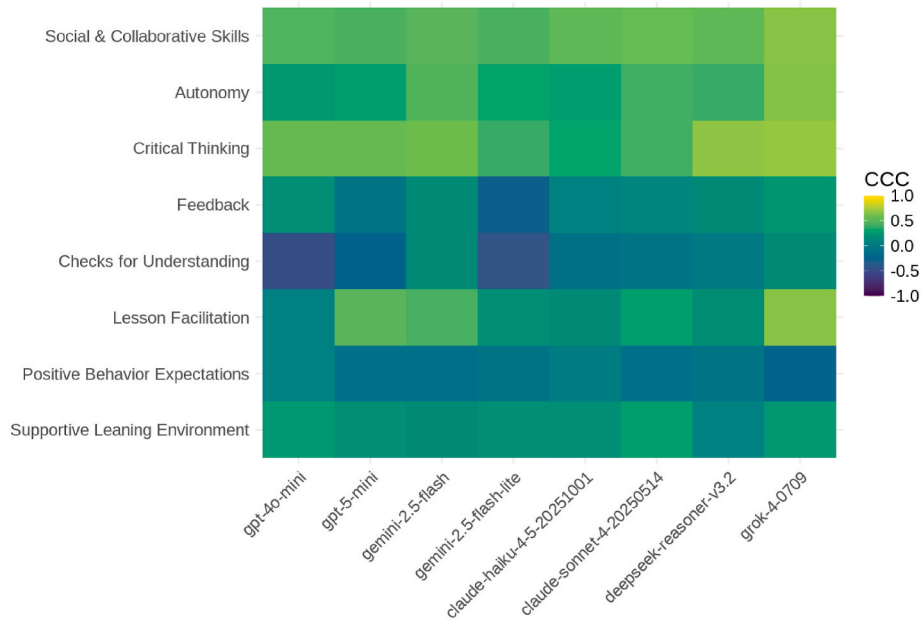


Fig. 2. Heatmap of mean Concordance Correlation Coefficient (CCC) by model and TEACH element. The element Perseverance is excluded from the figure because the expert ratings show zero variance, which makes CCC undefined.

Table 6

EA (mean) indicates the proportion of exact matches with expert ratings. MAE (mean) and RMSE (mean) report average errors across models and videos for each element. CCC (median) summarizes typical model-expert concordance, while “N Models with CCC < 0” indicates how many models produced negative concordance. CCC is not reported for Perseverance due to zero variance in expert ratings.

element	EA (mean)	MAE (mean)	RMSE (mean)	CCC (median)	N Models with CCC < 0
Supportive Learning Environment	0.46	0.69	0.99	0.19	0
Positive Behavior Expectations	0.24	1.14	1.4	−0.08	6/8
Lesson Facilitation	0.47	0.64	0.87	0.25	0
Checks for Understanding	0.27	1.12	1.46	−0.06	6/8
Feedback	0.28	0.99	1.25	0.12	2/8
Critical Thinking	0.47	0.65	0.94	0.55	0
Autonomy	0.38	0.74	0.99	0.36	0
Perseverance	0.69	0.33	0.55	-	-
Social & Collaborative Skills	0.56	0.67	1.11	0.51	0

lowest EA, the highest MAE and RMSE values, and are the only elements with negative median CCC. These two elements also show negative concordance for six of the eight evaluated models, in contrast to the remaining dimensions, which exhibit no negative CCC values. Overall, the results indicate that accuracy varies more sharply across instructional elements than across models, with certain pedagogical dimensions presenting persistent challenges for most systems.

4.3. Dimension-level differences in model reliability and accuracy

Fig. 3 displays the joint distribution of intra-model consistency (ICC) and accuracy relative to the expert ratings (CCC) for each model-element combination. The figure shows that most points are concentrated in the lower-left and lower-right regions, indicating that combinations tend to fall below the thresholds for “good” consistency and

“good” accuracy. No model-element pair reaches the upper-right quadrant, where both ICC and CCC exceed 0.75. Across elements, *Critical Thinking* stands out with several points located in the upper portion of the plot, reflecting comparatively higher CCC values across models. In contrast, *Checks for Understanding* shows multiple points with low CCC, several of them negative, distributed mostly below the horizontal threshold. From the perspective of models, *grok-4-0709* displays the largest number of combinations positioned toward the upper region of the figure, reflecting higher CCC values and several ICC values above the vertical threshold. Conversely, several combinations involving *gpt-4o-mini*, *gemini-2.5-flash-lite*, and *deepseek-reasoner-v3.2* appear in the lower parts of the graph, including the combination *Checks for Understanding-gpt-4o-mini*, which corresponds to the lowest CCC value in the dataset.

5. Discussion

This study set out to explore whether LLMs can approximate expert judgment in general, the study investigated how consistently they do so (RQ1), how closely their scores align with expert benchmarks (RQ2), and whether performance varies systematically across pedagogical dimensions (RQ3). We will discuss findings and their implication for the field, considering research and methodological consequences and pedagogical considerations.

5.1. The accuracy-reliability trade-off

Findings related to RQ1 demonstrate substantial intra-model variability across all evaluated LLMs, confirming concerns raised in prior work regarding the stochastic nature of generative models (Bauer et al., 2025; Seo et al., 2025). This variability is not incidental but reflects fundamental properties of LLM operation. Sampling-based decoding mechanisms—such as temperature scaling, nucleus sampling, and top-k filtering—introduce controlled randomness into token generation, and even under default or low-temperature settings, outputs remain non-deterministic due to probabilistic inference processes (Li et al., 2025). While this stochasticity is beneficial for open-ended generation tasks, it poses challenges in evaluative contexts where stability across repeated assessments is critical.

Importantly, variability in generated text does not remain confined

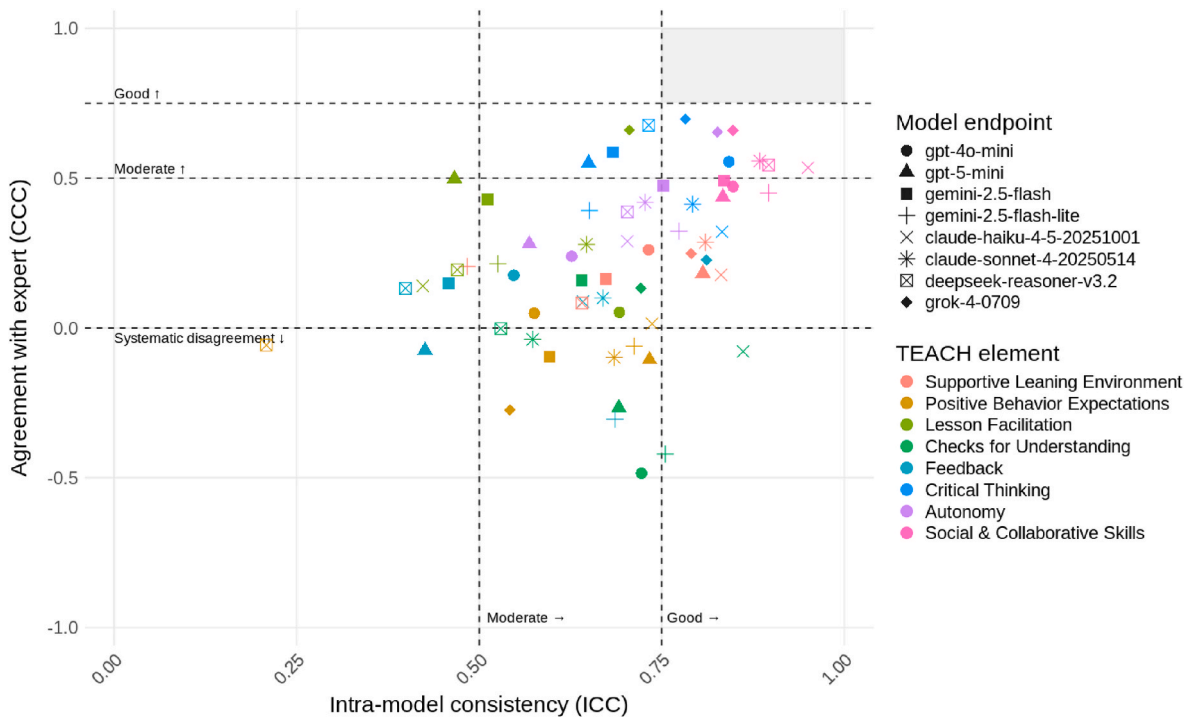


Fig. 3. Joint distribution of consistency (ICC) and accuracy (CCC) across TEACH elements and models.

to surface-level phrasing. In rubric-based evaluation tasks, small internal fluctuations can alter how models prioritize instructional evidence or interpret scoring criteria, leading to substantively different judgments. As observed in this study, identical prompts and transcripts can yield different element-level scores across repetitions, reflecting shifts in how the model weighs missing cues, isolated instructional behaviors, or partial evidence within the same lesson segment.

A second contributor to instability relates to attention allocation across long and instruction-dense prompts. Although modern LLMs support extended context windows, attention is not uniformly distributed, and small stochastic differences can cause models to focus on different rubric components across repetitions. In this study, evaluation prompts combined extensive rubric definitions and behavioral descriptors with full classroom transcripts, creating complex attention demands.

For example, evidence of attention-driven inconsistency emerges when comparing two evaluations generated by *gpt-4o-mini* from the same transcript for the same behavioral indicator (9.3, collaborative behavior). In one iteration, the model interpreted the transcript where the teacher said “three students come forward ... each chooses a group of bodies to organize” as a division of labor, concluding that interaction was insufficient for collaboration. In another repetition, the same episode was framed as collective action toward a shared organizational goal, warranting positive points for collaboration. The model's evaluative focus alternated between individualized student actions and the broader collective purpose of the task, producing contradictory judgements from identical evidence.

Divergences in attention focus, across repetitions, are consistent with prior research showing LLM outputs are highly sensitive to how attention is allocated across long and complex prompts. AlGhamdi et al. (2026) showed that modest changes in prompt structure and task sequencing significantly affected both scoring performance and feedback generation, while Song et al. (2024) reported performance degradation under long-context conditions due to model “forgetting”. Together, these findings suggest that prompt length and structural complexity can amplify variability in evaluative tasks.

Addressing RQ2, comparisons with the expert ratings revealed

moderate alignment for some model-element combinations, but no model reached more than 55% level of exact agreement. Models exhibiting lower dispersion did not consistently achieve stronger alignment with expert judgments, while models with higher agreement often displayed greater stochastic variability. For example, claude-haiku-4-5-20251001 exhibited the lowest dispersion across repetitions but also among the weakest alignment with expert ratings, while grok-4-0709 achieved the strongest agreement with expert judgment while displaying higher stochastic variability, underscoring that reliability and accuracy are not interchangeable properties in LLM-based classroom observation.

This decoupling extends concerns raised in the broader literature on LLM-as-a-judge systems, which has largely emphasized accuracy while giving limited attention to stability under repetition (Demszky et al., 2021; Li et al., 2025). From a measurement perspective, these findings reinforce the principle that reliability is a prerequisite for valid interpretation. In classroom observation contexts—where scores are intended to guide reflection or professional growth—unstable evaluations risk undermining both interpretability and trust, regardless of their average proximity to expert judgment.

5.2. Where pedagogical judgment resists automation: element-level constraints of LLMs in classroom observation

Results addressing RQ3 indicate that alignment between LLM-generated scores and expert ratings varies systematically across pedagogical dimensions. *Positive Behavior Expectations* exhibits the weakest performance, characterized by low exact agreement, high error, and negative concordance across most evaluated models. In contrast, elements such as *Critical Thinking* and *Social & Collaborative Skills* show substantially stronger alignment with expert judgment, despite similarly requiring interpretive judgment about classroom practice.

Closer examination of LLM-generated justifications helps explain these element-level discrepancies in pedagogical terms. For elements such as *Positive Behavior Expectations*, the TEACH Primary manual explicitly allows high ratings under two alternative conditions: either the teacher clearly sets behavioral expectations throughout the lesson,

or such expectations are not directly observed but students remain consistently well behaved. Correct application of this criterion therefore requires evaluators to recognize that the absence of explicit teacher actions does not necessarily indicate weak practice when sustained student self-regulation is evident.

To illustrate this pattern, we present an example from a single classroom video evaluated using the gemini-2.5-flash model, contrasting the model's justification with that reflected in the consensus-based expert ratings. In this case, the model assigned a low score, arguing that *"the teacher does not establish clear behavioral expectations for classroom tasks or activities."* The justification emphasized that instructions were limited to task completion—for example, when introducing an activity, *"only task-related instructions are provided,"* and similarly for individual work, *"instructions focus exclusively on completing the task, without mentioning behavioral expectations such as 'work quietly' or 'raise your hand if you have questions.'"* The consensus-based expert ratings, by contrast, explicitly applied the rubric's alternative condition, noting that *"the teacher does not establish clear behavioral expectations, but students demonstrate appropriate behavior throughout the class, which implies that expectations were established previously."*

This illustrative contrast reflects a broader pattern observed across models. When explicit behavioral directives are absent from the transcript, LLMs tend to interpret non-observation as insufficient evidence and default to lower scores. Certified observers, in contrast, apply the rubric's conditional logic by recognizing sustained student self-regulation as valid pedagogical evidence. This systematic bias toward evidentiary explicitness—rather than random inconsistency—helps explain the persistently low alignment observed for absence-tolerant elements such as *Positive Behavior Expectations*.

Taken together, these findings suggest that the observed limitations are not merely technical shortcomings of individual models, but pedagogical constraints rooted in how instructional quality is interpreted and evidenced within established observation frameworks. Classroom observation instruments such as TEACH embed professional judgment developed through training and experience, particularly in interpreting implicit signals and conditional evidence. When these judgments are operationalized through text-only inputs, LLMs tend to privilege explicit verbalization over contextual interpretation, revealing a gap between formal rubric definitions and expert pedagogical reasoning.

These results underscore the continued importance of pedagogical expertise in AI-assisted classroom observation. Effective use of LLMs depends not only on model capability, but on how their outputs are mediated, interpreted, and situated within the logic of the observation framework. In elements where meaning is inferred from patterns of student behavior rather than explicit teacher actions, expert judgment remains central. Looking forward, these findings point toward the need for hybrid human-AI systems and multimodal approaches that better approximate the evidentiary conditions under which professional observers operate, ensuring that automation supports—rather than substitutes—pedagogical judgment.

5.3. Implications for AI-assisted teacher professional development

The findings of this study suggest that the role of AI in teacher professional development is inherently conditional and closely tied to the nature of the pedagogical dimensions being evaluated. While large language models can generate timely and structured feedback, their effectiveness depends on the type of instructional evidence that becomes visible and actionable within the data representations used. In dimensions grounded in explicit and observable classroom behaviors, AI-generated feedback can support teacher noticing and reflection by highlighting patterns in practice and prompting focused analysis. In this sense, LLM-based systems may function as accessible diagnostic tools that complement reflective processes central to professional learning.

However, the results also raise important concerns regarding the use of AI-generated feedback in teacher self-assessment contexts, where

expert mediation may be limited or absent. The observed variability and systematic biases suggest that such uses may be pedagogically risky. When feedback is inconsistent across repeated evaluations or biased toward explicitly verbalized evidence, teachers engaging in AI-mediated self-assessment may receive contradictory or incomplete interpretations of their practice. In the absence of expert mediation, there is limited capacity to identify when such feedback is misleading. As a result, AI-supported self-assessment may not only fail to support professional growth, but may also lead to incorrect inferences about instructional quality. This risk is amplified when teachers engage with such feedback uncritically or without sufficient support to integrate it into their professional reasoning.

These risks are particularly evident in pedagogical dimensions that rely on the interpretation of absence or implicit classroom dynamics, especially when teachers engage with AI-generated feedback to inform their own practice. As shown in the analysis, LLMs tend to privilege explicit instructional cues while underrepresenting forms of evidence that expert observers recognize as valid, such as sustained student self-regulation. This limitation is compounded by the absence of non-verbal information in text-based representations, including gestures, eye contact, tone of voice, and spatial positioning, which are central to effective teaching. Research has shown that such non-verbal behaviors—often described as teacher immediacy—play a critical role in enhancing student motivation and engagement (Liu, 2021), and are also key channels through which teachers communicate expectations, emotions, and even implicit biases (Inan-Kaya & Rubie-Davis, 2022). As a result, AI-generated feedback may systematically foreground explicit, textually recoverable features while backgrounding these dimensions of practice. In turn, this may lead teachers to prioritize more explicit, verbalized instructional behaviors—those more readily captured by the model—while neglecting relational and embodied aspects of teaching that are less visible in textual data but pedagogically significant. Consequently, reliance on such feedback may inadvertently promote a narrower and more procedural conception of teaching, potentially shaping practice in ways that are misaligned with established evidence on effective classroom interaction.

Taken together, these findings underscore that the effective use of AI in teacher professional development depends not only on model performance, but on the broader professional and relational conditions under which feedback is interpreted and integrated into reflective processes. AI-generated outputs should not be treated as self-sufficient evaluations, but as inputs requiring pedagogical mediation. Without such mediation, there is a risk that reflection becomes oriented toward what is easily measurable or explicitly represented, rather than what is instructionally meaningful. This may lead teachers to prioritize observable behaviors—such as the explicit verbalization of expectations—over deeper instructional intentions and relational dimensions of teaching, resulting in forms of professional learning that are more reactive than reflective.

Furthermore, the findings suggest that AI-mediated feedback may influence how teachers conceptualize effective teaching. By systematically privileging explicit and textually observable behaviors, these systems risk reinforcing a narrowed view of instructional quality that underrepresents tacit knowledge, embodied interaction, and context-sensitive judgment, making certain dimensions of teaching more visible while rendering others effectively invisible within evaluation processes. In the absence of strong interpretive frameworks, teachers may adopt AI-generated feedback uncritically, increasing the likelihood of misalignment between feedback and effective practice. Supporting critical engagement with such feedback is therefore essential to ensure that AI contributes meaningfully to teacher professional development.

6. Conclusion

This study examined the use of LLMs as automated classroom observers within a high-inference instructional framework, focusing on

reliability, accuracy, and element-level validity. The findings contribute to a growing understanding of AI in education as a context-sensitive and task-dependent technology rather than a universally applicable solution.

From a professional development perspective, the results indicate that the value of AI-assisted classroom observation is conditional. While automated feedback can support reflection in dimensions grounded in explicit instructional behaviors, its use in high-inference or absence-based elements remains limited. In such cases, the lack of alignment with expert judgment introduces risks of misinterpretation, particularly when feedback is used without appropriate pedagogical mediation. This highlights the continued importance of expert judgment in interpreting and contextualizing AI-generated evaluations.

A central implication of this work is that validation must precede scale not only in deployment, but also in research design. Studies evaluating AI-assisted classroom observation should explicitly account for repeatability and intra-model variability, rather than relying on single-pass evaluations. Without repeated-measures analysis, reported accuracy estimates may obscure substantial stochastic instability and overstate model reliability. From a methodological perspective, element-level validation, systematic repetition of identical evaluation conditions, and benchmarking against certified expert judgments should be treated as minimum requirements for credible evidence claims in this domain.

Consistent with prior calls for hybrid approaches (Jacobs et al., 2025; Wang, Gong, et al., 2025), the findings support a model of human-AI complementarity rather than substitution. Progress in AI-assisted classroom observation is therefore likely to depend on three interconnected directions. First, human-AI collaboration models can leverage computational efficiency and pattern detection while preserving expert oversight for contextual interpretation and professional judgment. Second, multimodal systems that integrate visual and auditory classroom cues offer a pathway to address limitations inherent in transcript-only analysis, particularly for regulatory and interactional dimensions of teaching. Third, agent-based architectures that explicitly model expert pedagogical knowledge beyond the procedural definitions in observation manuals may help bridge the gap between formal rubrics and professional reasoning practices. Together, these directions point toward hybrid system designs that combine computational scalability with pedagogical and measurement integrity, rather than pursuing full automation as an end in itself.

Several limitations of this study point toward important directions for continued research. First, the dataset consisted of 12 classroom recordings. Although these were drawn from standardized training materials and evaluated through extensive repetition, larger and more diverse samples are needed to assess the robustness of the findings across instructional contexts, grade levels, and subject domains. Second, while the inclusion of English- and Spanish-language videos introduces linguistic and contextual variation, the TEACH framework itself is designed for cross-cultural application. Observed differences may therefore reflect language-specific model performance rather than limitations of the observation instrument. Future studies should examine how multilingual model training and language-specific representations influence evaluative accuracy. Third, although repeated evaluations were used to estimate intra-model variability, the number of repetitions was not intended to produce highly precise performance estimates for individual models. As such, the results should not be interpreted as definitive benchmarks or rankings across endpoints. Finally, the inherent asymmetry between the experts' evaluation conditions and those available to the LLMs introduces an important consideration for interpreting the results. While expert ratings were based on full video recordings, models operated on transcripts alone, reflecting realistic usage conditions. As a result, observed differences between LLM and expert evaluations may be influenced not only by model capability, but also by the absence of multimodal information in the input. This design choice enables the study of AI-assisted observation under practical constraints, but it also limits the extent to which discrepancies can be attributed solely to the

models themselves.

Future research should build on these findings by moving beyond score-based evaluation toward a deeper analysis of the feedback generated by AI systems, including how explanations are constructed and how they influence teacher interpretation and learning. In addition, studies should examine the direct impact of AI-assisted feedback on teacher practices and student outcomes, particularly in authentic professional development settings. Further work is also needed to explore approaches that incorporate expert pedagogical knowledge more explicitly into AI systems, including agent-based or knowledge-grounded designs that extend beyond procedural rubric definitions. An important direction is to situate LLM performance relative to human inter-rater variability, in order to better contextualize observed levels of agreement and stability. Finally, the development and evaluation of multimodal models represent a critical direction for capturing the full complexity of classroom interaction and addressing limitations identified in text-based analysis.

CRedit authorship contribution statement

Carolina Melo: Writing – original draft, Supervision, Resources, Conceptualization. **Javiera de la Maza:** Writing – original draft, Validation, Data curation. **Matías Recabarren:** Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization.

Statements on open data and ethics

The underlying classroom video recordings and transcripts are not publicly available, as they are derived from proprietary training materials owned and managed by the World Bank and are subject to access restrictions. Derived analytical outputs and code used for statistical analyses may be made available from the corresponding author upon reasonable request, subject to permissions and data-sharing restrictions.

This study uses secondary data provided through the TEACH program. The authors did not collect primary data nor have direct contact with participants. All materials were used in accordance with the conditions established by the data providers, and analyses were conducted on de-identified data. The study focuses on evaluating AI systems, and all results are reported in aggregate form.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with language refinement and editorial improvements. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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